Appendix A: Equivalence Validation - TSENAT vs SplicingFactory

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Introduction

This appendix validates that **TSENAT's transcript diversity measures (Shannon entropy, Simpson index) are mathematically equivalent to SplicingFactory's implementations**. We benchmark both packages on TCGA BRCA RNA-seq data to demonstrate:

- Shannon equivalence: TSENAT Tsallis q=1 ~ SplicingFactory Shannon
- Simpson equivalence: TSENAT Tsallis q=2 ~ SplicingFactory Simpson

- Identical statistical results: Same gene rankings, p-values, fold changes

Who needs this appendix? Anyone evaluating TSENAT for transcript diversity analysis or validating cross-package comparisons.

Benchmarking

For background on transcript diversity measures and the Tsallis entropy framework, see the main vignette. We use the TCGA BRCA RNA-seq dataset included with the SplicingFactory package as our validation benchmark.

Importing example data

We begin by loading the required libraries and the TCGA BRCA dataset that will serve as our validation benchmark.

```
suppressPackageStartupMessages({
    library("SplicingFactory")
    library(TSENAT)
    library("SummarizedExperiment")
})
set.seed(12345)

# Load dataset
data(tcga_brca_luma_dataset)

# Extract gene names
```

```r
genes <- tcga_brca_luma_dataset[, 1]

# Extract read count data without gene names
readcounts <- tcga_brca_luma_dataset[, -1]
```

Data filtering and preprocessing

We filter genes to retain only those with sufficient coverage across samples, ensuring robust diversity estimates.

```r
tokeep <- rowSums(readcounts > 5) > 5
readcounts <- readcounts[tokeep, ]
genes      <- genes[tokeep]

# Create SummarizedExperiment object for cleaner data handling
# Important: Set gene names as rownames for compatibility with SplicingFactory
# TSENATAnalysis requires 'gene_id' in rowData
se_data <- SummarizedExperiment(
  assays = list(counts = as.matrix(readcounts)),
  rowData = DataFrame(gene_id = genes)
)
rownames(se_data) <- genes
```

Transcript diversity calculation

To calculate transcript diversity measures (Shannon entropy, Simpson index, and Tsallis entropy at q=1 and q=2), use:

```r
# Create sample group classification based on sample names
# (Normal samples end with _N, Tumor samples don't)
sample_group <- ifelse(grepl("_N$", colnames(se_data)), "Normal", "Tumor")

# Add group metadata to se_data for all downstream analyses
colData(se_data)$group <- sample_group

# SplicingFactory: Calculate diversity for both Shannon (naive) and Simpson methods
sf_methods <- c("naive", "simpson")
splicing_results <- setNames(vector("list", length(sf_methods)), sf_methods)

for (method in sf_methods) {
  splicing_results[[method]] <- SplicingFactory::calculate_diversity(
    x = se_data,
    method = method,
    norm = TRUE,
    verbose = FALSE
  )
  # Add sample_type and group metadata to SplicingFactory objects
  # (sample_type matches SplicingFactory naming; group matches TSENAT naming for consistency)
  colData(splicing_results[[method]])$sample_type <- sample_group
  colData(splicing_results[[method]])$group <- sample_group
}
```

```
# TSENAT: Single config for q-value sweep
# (Configuration is identical for all q-values)
config <- TSENAT_config(condition_col = "group")

# TSENAT: Create analysis objects and compute diversity for q=1 and q=2
q_values <- c(1, 2)
tsenat_results <- setNames(vector("list", length(q_values)), paste0("q", q_values))

for (q in q_values) {
  # Create analysis object (following Bioconductor pattern: config first, then constructor)
  tsenat_results[[paste0("q", q)]] <- TSENATAnalysis(se_data, config = config)

  # Compute diversity with dynamic output filename using sprintf
  # Use explicit namespace to avoid SplicingFactory::calculate_diversity masking
  tsenat_results[[paste0("q", q)]] <- TSENAT::calculate_diversity(
    analysis = tsenat_results[[paste0("q", q)]],
    q = q,
    norm = TRUE,
    verbose = FALSE
  )
}

# Extract for downstream use (aliases maintain backward compatibility with existing code)
shannon_entropy <- splicing_results$naive
simpson_index <- splicing_results$simpson
tsenat_analysis_q1 <- tsenat_results$q1
tsenat_analysis_q2 <- tsenat_results$q2
```

Both packages return a `SummarizedExperiment` object, that you can investigate further with the `assay` function.

Differential analysis

We apply statistical testing to identify genes with significantly different transcript diversity between Normal and Tumor samples using both SplicingFactory and TSENAT frameworks.

```
# SplicingFactory: Differential analysis for Shannon entropy
entropy_significance <- SplicingFactory::calculate_difference(
  x = shannon_entropy,
  samples = "sample_type",
  control = "Normal",
  method = "mean",
  test = "wilcoxon",
  verbose = FALSE
)

# SplicingFactory: Differential analysis for Simpson index
simpson_significance <- SplicingFactory::calculate_difference(
  x = simpson_index,
  samples = "sample_type",
  control = "Normal",
  method = "mean",
```

```r
  test = "wilcoxon",
  verbose = FALSE
)


# TSENAT: Differential analysis for Tsallis q=1 (Shannon equivalent)
# Extract the diversity results for q=1 using the results() accessor
# Returns SummarizedExperiment with diversity values (genes x samples)
div_result_q1 <- results(tsenat_analysis_q1, type = "diversity", q = 1.0, format = "se")

# Apply SplicingFactory's calculate_difference to TSENAT q=1 diversity
# Use "group" column already in colData (from se_data)
tsenat_shannon_diff <- SplicingFactory::calculate_difference(
  x = div_result_q1,
  samples = "group",
  control = "Normal",
  method = "mean",
  test = "wilcoxon",
  verbose = FALSE
)


# TSENAT: Differential analysis for Tsallis q=2 (Simpson equivalent)
# Extract the diversity results for q=2 using the results() accessor
# Returns SummarizedExperiment with diversity values (genes x samples)
div_result_q2 <- results(tsenat_analysis_q2, type = "diversity", q = 2.0, format = "se")

# Apply SplicingFactory's calculate_difference to TSENAT q=2 diversity
# Use "group" column already in colData (from se_data)
tsenat_simpson_diff <- SplicingFactory::calculate_difference(
  x = div_result_q2,
  samples = "group",
  control = "Normal",
  method = "mean",
  test = "wilcoxon",
  verbose = FALSE
)
```

For the benchmark comparison with TSENAT, we will compute Tsallis entropy at q=1 (equivalent to Shannon) and q=2 (equivalent to Simpson). This allows direct comparison with the SplicingFactory results shown above.

Benchmark Results Summary

Shannon Entropy (SplicingFactory Laplace vs TSENAT Tsallis q=1)

Table 1: Supplementary Table 8 | Shannon Entropy Analysis Summary Statistics. Differential analysis comparing Normal (N=8) vs Tumor (N=8) samples using SplicingFactory's `calculate_difference()` method.

Method	Genes Tested	Significant (padj<0.05)	Mean log2FC	SD log2FC
SplicingFactory (Shannon/naive)	214	26	-0.0009354	1.582967
TSENAT (Tsallis q=1)	213	26	0.0003618	1.586675

Interpretation: Both methods similarly identify significantly different transcript diversity between Normal and Tumor samples. The comparable number of significant genes (padj<0.05) and mean log2FC values demonstrate equivalent statistical power and effect size detection.

Top 10 Significant Genes - SplicingFactory (Laplace)

Table 2: **Supplementary Table 9 | SplicingFactory method - Top 10 genes identified by Shannon entropy (naive mode).** Ranked by adjusted *P*-value.

Gene	Normal Mean	Tumor Mean	Mean Diff	log2FC	p-value	adj p-value
C1orf213	0.8122	0.1400	-0.6722	-2.5365	3.97e-07	8.50e-05
HAPLN3	0.7030	0.4299	-0.2731	-0.7096	4.17e-05	4.46e-03
COL1A2	0.0000	0.1643	0.1643	Inf	6.68e-05	4.77e-03
F10	0.1311	0.3720	0.2409	1.5050	1.37e-04	7.31e-03
DUSP14	0.2989	0.1302	-0.1688	-1.1995	1.89e-04	8.10e-03
MBD2	0.0027	0.0371	0.0344	3.7876	2.39e-04	8.54e-03
C1orf86	0.1479	0.0029	-0.1450	-5.6777	4.59e-04	9.85e-03
GFPT1	0.0231	0.0025	-0.0207	-3.2253	3.80e-04	9.85e-03
HNRNPR	0.5162	0.5907	0.0746	0.1947	4.60e-04	9.85e-03
OSR1	0.0497	0.2891	0.2394	2.5411	4.10e-04	9.85e-03

Top 10 Significant Genes - TSENAT (Tsallis q=1)

Table 3: Supplementary Table 10 | TSENAT method - Top 10 genes identified by Tsallis entropy at q=1 (Shannon equivalence). Results obtained using SplicingFactory's `calculate_difference()` method applied to TSENAT Tsallis q=1 diversity values. Ranked by adjusted p-value.

Gene	Normal Mean	Tumor Mean	Mean Diff	log2FC	p-value	adj p-value
C1orf213	0.8122	0.1400	-0.6722	-2.5365	3.97e-07	8.46e-05
HAPLN3	0.7030	0.4299	-0.2731	-0.7096	4.17e-05	4.44e-03
COL1A2	0.0000	0.1643	0.1643	Inf	6.68e-05	4.74e-03
F10	0.1311	0.3720	0.2409	1.5050	1.37e-04	7.27e-03
DUSP14	0.2989	0.1302	-0.1688	-1.1995	1.89e-04	8.06e-03
MBD2	0.0027	0.0371	0.0344	3.7876	2.39e-04	8.50e-03
C1orf86	0.1479	0.0029	-0.1450	-5.6777	4.59e-04	9.80e-03
HNRNPR	0.5162	0.5907	0.0746	0.1947	4.60e-04	9.80e-03
OSR1	0.0497	0.2891	0.2394	2.5411	4.10e-04	9.80e-03

Gene	Normal Mean	Tumor Mean	Mean Diff	log2FC	p-value	adj p-value
GFPT1	0.0231	0.0025	-0.0207	-3.2253	3.80e-04	9.80e-03

Simpson Index (SplicingFactory vs TSENAT Tsallis q=2)

Table 4: Supplementary Table 11 | Simpson Index Analysis Summary Statistics. Differential analysis using Simpson diversity metric (Gini-Simpson index) comparing Normal (N=8) vs Tumor (N=8) samples using SplicingFactory's `calculate_difference()` method.

Method	Genes Tested	Significant (padj<0.05)	Mean log2FC	SD log2FC
SplicingFactory (Simpson)	214	26	-0.0012687	1.812772
TSENAT (Tsallis q=2)	213	26	-0.0000521	1.817060

Interpretation: Simpson index results parallel Shannon entropy findings, with both methods identifying similar numbers of significantly different isoform dominance patterns. The agreement across both diversity measures provides strong evidence for TSENAT's mathematical and statistical equivalence to SplicingFactory.

Top 10 Significant Genes - SplicingFactory (Simpson)

Table 5: **Supplementary Table 12 | SplicingFactory method - Top 10 genes by Simpson index (Gini-Simpson, emphasizes dominant isoforms).** Ranked by adjusted *P*-value.

Gene	Normal Mean	Tumor Mean	Mean Diff	log2FC	p-value	adj p-value
C1orf213	0.3852	0.0515	-0.3337	-2.9037	3.97e-07	8.50e-05
HAPLN3	0.4823	0.2558	-0.2266	-0.9152	5.17e-06	5.53e-04
COL1A2	0.0000	0.0626	0.0626	Inf	6.68e-05	4.77e-03
F10	0.0697	0.2494	0.1797	1.8400	1.10e-04	5.86e-03
DUSP14	0.1686	0.0617	-0.1069	-1.4499	2.11e-04	7.32e-03
HNRNPR	0.5376	0.6134	0.0758	0.1903	1.79e-04	7.32e-03
MBD2	0.0005	0.0107	0.0102	4.5028	2.39e-04	7.32e-03
CXorf40A	0.7455	0.7939	0.0484	0.0907	2.75e-04	7.34e-03
GFPT1	0.0051	0.0004	-0.0047	-3.5271	3.80e-04	8.78e-03
OSR1	0.0124	0.1156	0.1032	3.2168	4.10e-04	8.78e-03

Top 10 Significant Genes - TSENAT (Tsallis q=2)

Table 6: Supplementary Table 13 | TSENAT method - Top 10 genes by Tsallis entropy at q=2 (Simpson equivalence). Results obtained using SplicingFactory's `calculate_difference()` method applied to TSENAT Tsallis q=2 diversity values. Ranked by adjusted p-value.

Gene	Normal Mean	Tumor Mean	Mean Diff	log2FC	p-value	adj p-value
C1orf213	0.7703	0.1029	-0.6674	-2.9037	3.97e-07	8.46e-05
HAPLN3	0.7235	0.3836	-0.3398	-0.9152	5.17e-06	5.50e-04
COL1A2	0.0000	0.1251	0.1251	Inf	6.68e-05	4.74e-03
F10	0.1045	0.3741	0.2696	1.8400	1.10e-04	5.83e-03
HNRNPR	0.6144	0.7010	0.0866	0.1903	1.79e-04	7.28e-03
DUSP14	0.2529	0.0926	-0.1603	-1.4499	2.11e-04	7.28e-03
MBD2	0.0009	0.0213	0.0204	4.5028	2.39e-04	7.28e-03
CXorf40A	0.8387	0.8931	0.0544	0.0907	2.75e-04	7.31e-03
OSR1	0.0249	0.2312	0.2063	3.2168	4.10e-04	8.74e-03
GFPT1	0.0103	0.0009	-0.0094	-3.5271	3.80e-04	8.74e-03

Analysis: Simpson Index and Tsallis Entropy at q=2

Both methods share the same mathematical formula:

$$Simpson = Tsallis_{q=2} = 1 - \sum_{i=1}^{n} p_i^2$$

where $p_i = \frac{x_i}{\sum_i x_i}$ (transcript proportions).

However, SplicingFactory returns raw values, while TSENAT normalizes by the theoretical maximum $(1 - 1/n)$:

	SplicingFactory	TSENAT
Formula	$1 - \sum p_i^2$	$\frac{1-\sum p_i^2}{1-1/n}$
Range	0 to ~1	0 to 1
Scale difference	Raw	~2x for 2-isoform genes

This explains observed differences like *C1orf213* Normal: 0.3852 (SplicingFactory) vs 0.7703 (TSENAT). Since C1orf213 has 2 expressed isoforms, the normalization factor is $1 - 1/2 = 0.5$. The calculation: $0.3852/0.5 = 0.7704$ confirms this scaling factor. Despite raw value differences, statistical analysis yields identical results:

1. log2 fold changes: Normalization factor cancels in ratios: $\log_2\left(\frac{S_{norm,t}}{S_{norm,n}}\right) = \log_2\left(\frac{S_{raw,t}/K}{S_{raw,n}/K}\right) = \log_2\left(\frac{S_{raw,t}}{S_{raw,n}}\right)$
2. P-values and rankings: Wilcoxon rank-sum tests depend only on gene ranks, not absolute values, so both methods identify the same significant genes
3. Statistical validity: Both approaches are correct-SplicingFactory favors unbounded diversity, TSENAT favors interpretable [0,1] scaling

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Conclusion: Equivalence Validation Summary

This appendix has demonstrated that TSENAT's implementation of transcript diversity measures is mathematically and statistically equivalent to SplicingFactory's established methods for both Shannon entropy and Simpson index calculations.

Session Information

```
sessionInfo()
#> R version 4.5.3 (2026-03-11)
#> Platform: x86_64-conda-linux-gnu
#> Running under: Ubuntu 22.04.5 LTS
#>
#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblasp-r0.3.30.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=es_ES.UTF-8       LC_NUMERIC=C
#>  [3] LC_TIME=de_DE.UTF-8        LC_COLLATE=es_ES.UTF-8
#>  [5] LC_MONETARY=de_DE.UTF-8    LC_MESSAGES=es_ES.UTF-8
#>  [7] LC_PAPER=de_DE.UTF-8       LC_NAME=C
#>  [9] LC_ADDRESS=C               LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats4    stats     graphics  grDevices utils     datasets  methods
#> [8] base
#>
#> other attached packages:
#>  [1] SplicingFactory_1.18.0      knitr_1.51
#>  [3] dplyr_1.2.1                 SummarizedExperiment_1.40.0
#>  [5] Biobase_2.70.0              GenomicRanges_1.62.1
#>  [7] Seqinfo_1.0.0               IRanges_2.44.0
#>  [9] S4Vectors_0.48.1            BiocGenerics_0.56.0
#> [11] generics_0.1.4              MatrixGenerics_1.22.0
#> [13] matrixStats_1.5.0           kableExtra_1.4.0
#> [15] BiocStyle_2.38.0            TSENAT_0.99.35
#> [17] testthat_3.3.2
#>
#> loaded via a namespace (and not attached):
#>  [1] Rdpack_2.6.6        sandwich_3.1-1      rlang_1.2.0
#>  [4] magrittr_2.0.5      multcomp_1.4-30     otel_0.2.0
#>  [7] compiler_4.5.3      mgcv_1.9-4          systemfonts_1.3.2
#> [10] vctrs_0.7.3         stringr_1.6.0       pkgconfig_2.0.3
#> [13] fastmap_1.2.0       backports_1.5.1     XVector_0.50.0
#> [16] rmarkdown_2.31      tzdb_0.5.0          nloptr_2.2.1
```

```
#> [19] tinytex_0.59        purrr_1.2.2         xfun_0.58
#> [22] cachem_1.1.0        jsonlite_2.0.0      DelayedArray_0.36.1
#> [25] BiocParallel_1.44.0 broom_1.0.13        parallel_4.5.3
#> [28] R6_2.6.1            stringi_1.8.7       bslib_0.11.0
#> [31] RColorBrewer_1.1-3  car_3.1-5           boot_1.3-32
#> [34] pkgload_1.5.2       brio_1.1.5          jquerylib_0.1.4
#> [37] estimability_1.5.1  bookdown_0.46       Rcpp_1.1.1-1.1
#> [40] zoo_1.8-15          readr_2.2.0         Matrix_1.7-5
#> [43] splines_4.5.3       tidyselect_1.2.1    rstudioapi_0.18.0
#> [46] abind_1.4-8         yaml_2.3.12         codetools_0.2-20
#> [49] pkgbuild_1.4.8      lattice_0.22-9      tibble_3.3.1
#> [52] plyr_1.8.9          withr_3.0.3         S7_0.2.2
#> [55] evaluate_1.0.5      desc_1.4.3          survival_3.8-6
#> [58] xml2_1.5.2          BiocManager_1.30.27 pillar_1.11.1
#> [61] carData_3.0-6       reformulas_0.4.4    rprojroot_2.1.1
#> [64] hms_1.1.4           ggplot2_4.0.3       scales_1.4.0
#> [67] minqa_1.2.8         ARTool_0.11.2       xtable_1.8-8
#> [70] glue_1.8.1          pheatmap_1.0.13     emmeans_2.0.3
#> [73] tools_4.5.3         lme4_2.0-1          mvtnorm_1.3-6
#> [76] cowplot_1.2.0       grid_4.5.3          tidyr_1.3.2
#> [79] rbibutils_2.4.1     nlme_3.1-169        Formula_1.2-5
#> [82] cli_3.6.6           textshaping_1.0.5   viridisLite_0.4.3
#> [85] S4Arrays_1.10.1     svglite_2.2.2       geepack_1.3.13
#> [88] gtable_0.3.6        sass_0.4.10         digest_0.6.39
#> [91] SparseArray_1.10.10 TH.data_1.1-5       farver_2.1.2
#> [94] memoise_2.0.1       htmltools_0.5.9     lifecycle_1.0.5
#> [97] MASS_7.3-65
```